

ABSTRACT

The present work investigates the design and analysis of a staircase lift, which can be used as a Material Handling System. A staircase lift is a mechanical device for lifting people and wheelchairs up and down on the stairs, who may find difficulty in doing so themselves. For sufficiently wide stairs, a rail is mounted to the treads of the stairs. A chair or lifting platform is attached to the rail. A person on the chair or platform is lifted as the chair or platform moves along the rail, old age and goods are to be carried across the staircase. Staircase lift is a type of lift that can be mounted on the staircase without altering civil structure. This lift runs on electric power and consists of a motor, reduction gear box, rope drive, two rails and a sliding chair. In this system we use DC motor for changing the polarity of the power supply which will make the motor run in reverse direction. Advantages over the conventional hydraulic lift are no civil structure and alteration is required, low cost, less bulkiness, less power, less maintenance requires. Easy design, easy installations can be of industrial use too. Moreover, considering some drawbacks due to weight carrying capacity completely depend upon the capacity of motor. There is lot of scope for further modification in the project as using monorail instead of two, use of belt drive or chain drive instead of rope drive, Incorporation and automation/ timer unit which will ease the use of device. Rack and carrier arrangement for using the device for curved staircase and use of work & roller reduction gear assembly.

Key words: Platform, Rails, Wire rope drive, AC motor, civil structure.